Statistical Arbitrage via Bivariate Cointegration and Multivariate Johansen Baskets: Mathematical Foundations and Walk-Forward Backtesting Architecture

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Abstract

This report presents the design, mathematical formulation, and empirical evaluation of a quantitative Statistical Arbitrage (Stat-Arb) system. Rather than attempting to predict the absolute future direction of individual stock prices—which behave approximately as non-stationary random walks—we focus on relative pricing anomalies within homogeneous market sectors. We implement and compare two distinct econometric approaches: the classical two-step Engle-Granger bivariate pairs trading model and the multivariate Johansen trace cointegration framework for 3-asset baskets. Particular emphasis is placed on bridging the gap between theoretical time-series econometrics and realistic portfolio execution. We detail our multi-stage statistical filtering cascade, address the multiple hypothesis testing problem (False Discovery Rate vs. structural filtering), and establish a strict Out-of-Sample (OOS) Walk-Forward validation engine that accounts for look-ahead bias, next-open execution, transaction commissions, bid-ask slippage, and short-selling borrowing fees.

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*Source code and backtesting engine available at: https://github.com/Patryniuuu/StatArb

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1 Introduction and Problem Statement

1.1 What is Statistical Arbitrage?

In classical financial mathematics, deterministic arbitrage refers to a risk-free profit opportunity arising from a price discrepancy between identical or payoff-equivalent assets in different markets. In modern equity markets, pure deterministic arbitrage is virtually non-existent for retail or standard quantitative participants due to high-frequency market makers.

Statistical Arbitrage (Stat-Arb) relaxes the requirement of a risk-free payoff. Instead, it treats arbitrage as a probabilistic, mean-reverting process. The core premise is simple: while individual stock log-prices $\ln(P_t)$ are non-stationary processes whose variance grows linearly with time, specific linear combinations of stocks operating within the same industry sector often share common macroeconomic drivers (such as raw material costs, interest rate sensitivity, or consumer demand). When the relative valuation (the *spread*) between such assets diverges significantly from its historical equilibrium, a Stat-Arb strategy simultaneously opens a long position in the undervalued asset(s) and a short position in the overvalued asset(s), betting that the spread will revert to its long-term mean.

1.2 Why Correlation is Not Enough: Correlation vs. Cointegration

A common pitfall in naive quantitative trading is confusing return correlation with price cointegration:

- **Return Correlation** measures the short-term co-movement of period-to-period log-returns $r_t = \ln(P_t) - \ln(P_{t-1})$. Two stocks can exhibit a high daily return correlation ($\rho > 0.70$) because both react similarly to daily market news, yet their actual price levels can slowly drift apart over months without ever converging again.
- **Cointegration** is a structural property of the price levels themselves. If two or more non-stationary time series are cointegrated, there exists a linear combination of their log-prices that is strictly stationary—meaning its mean and variance remain constant over time. Cointegration acts as an invisible elastic band pulling the assets back together whenever they drift apart.

1.3 Scope and Objectives of This Project

The goal of this project is to build a complete, end-to-end quantitative research and backtesting pipeline from scratch, avoiding black-box backtesting libraries in order to maintain full mathematical control over every step. Specifically, our system investigates:

1. **Bivariate vs. Multivariate Structures:** Comparing 2-legged pairs (Engle-Granger) against 3-legged synthetic baskets (Johansen) across 36 liquid US equities divided into four homogeneous sectors (Technology/Semiconductors, Banks/Finance, Energy/Oil, and Consumer Staples).
2. **Statistical Quality Control:** Designing a multi-stage filtering cascade that eliminates spurious regressions, unbalanced hedge ratios, and one-way price drifts.
3. **Realistic Market Friction and Time-Scale Sensitivity:** Evaluating how transaction costs, short-borrowing fees, and execution delays (*Next Open* execution) impact strategy profitability across both daily (`1d`) and intraday hourly (`1h`) intervals.

2 Mathematical and Statistical Foundations

2.1 Order of Integration and Log-Price Representation

We model asset prices using natural logarithms, denoted as $y_t = \ln(P_{Y,t})$ and $x_t = \ln(P_{X,t})$. Working in log-space ensures that price changes are additive and correspond to continuously compounded percentage returns, making the regression slope β directly interpretable as a capital elasticity (dollar hedge ratio).

A discrete time series z_t is said to be integrated of order d, denoted $z_t \sim I(d)$, if it requires d differencing operations to become weakly stationary. Financial log-prices are empirically modeled as $I(1)$ processes (random walks with drift):

$$y_t = y_{t-1} + \mu + \eta_t, \quad \eta_t \sim \text{i.i.d. } (0, \sigma_\eta^2) \tag{1}$$

While $y_t \sim I(1)$ and $x_t \sim I(1)$ individually wander without bound, the pair (y_t, x_t) is cointegrated of order $CI(1,1)$ if there exists a cointegrating vector $[1, -\beta]^T$ such that the linear combination:

$$\varepsilon_t = y_t - \alpha - \beta x_t \sim I(0) \tag{2}$$

is a stationary process with constant mean and finite, time-invariant variance.

2.2 The Engle-Granger Two-Step Method (Bivariate Pairs)

For our 2-legged Pairs Trading strategy, we implement the Engle-Granger procedure.

2.2.1 Step 1: Ordinary Least Squares (OLS) Estimation

First, we assign the asset with the larger market capitalization (or lower return volatility if market cap data is unavailable) as the independent explanatory variable x_t, and the smaller asset as the dependent variable y_t. Given a training sample of T observations, we construct the design matrix $\mathbf{X} \in \mathbb{R}^{T \times 2}$ with a leading column of ones and a column of x_t:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_T \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_T \end{bmatrix} \tag{3}$$

The analytical OLS estimator $\hat{\mathbf{B}} = [\hat{\alpha}, \hat{\beta}]^T$ is computed directly via matrix inversion:

$$\hat{\mathbf{B}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{4}$$

From the resulting residual vector $\hat{\varepsilon}_t = y_t - (\hat{\alpha} + \hat{\beta}x_t)$, we also compute the coefficient of determination R^2 directly from the residual sum of squares (SS_{res}) and total sum of squares (SS_{tot}):

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{t=1}^{T} \hat{\varepsilon}_t^2}{\sum_{t=1}^{T} (y_t - \bar{y})^2} \tag{5}$$

2.2.2 Step 2: Augmented Dickey-Fuller (ADF) Test with MacKinnon Critical Values

To verify whether the estimated spread $\hat{\varepsilon}_t$ is stationary ($I(0)$), we apply the Augmented Dickey-Fuller test. Since the OLS residuals have a sample mean of zero by construction ($\bar{\varepsilon} = 0$), we estimate the ADF regression without a constant term (`regression='n'`):

$$\Delta\hat{\varepsilon}_t = \gamma\hat{\varepsilon}_{t-1} + \sum_{i=1}^{p} \phi_i \Delta\hat{\varepsilon}_{t-i} + u_t \tag{6}$$

where the lag order p is selected automatically via the Akaike Information Criterion (AIC). We test the null hypothesis of a unit root ($H_0 : \gamma = 0$, non-stationarity) against the one-sided alternative of stationarity ($H_1 : \gamma < 0$) using the t-statistic:

$$t_{\text{ADF}} = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})} \tag{7}$$

Crucial Statistical Correction: Because $\hat{\varepsilon}_t$ is not an observed series but a series of residuals generated by minimizing variance in an OLS regression with two variables and an intercept, the standard Dickey-Fuller distribution is invalid (it is biased toward rejecting H_0 too often). Therefore, we evaluate t_{ADF} using MacKinnon's response surface p-values calibrated specifically for cointegration with $N = 2$ variables and a constant trend (`mackinnonp(t_stat, regression='c', N=2)`). A pair passes the stationarity check if $p_{\text{MacKinnon}} \leq 0.05$.

2.3 Ornstein-Uhlenbeck Process and Half-Life of Mean Reversion

Even if a spread is statistically stationary, its speed of mean reversion might be too slow to be profitable over a quarterly trading window. We model the continuous-time dynamics of the spread S_t as an Ornstein-Uhlenbeck (OU) stochastic differential equation:

$$dS_t = \lambda(\mu - S_t)dt + \sigma dW_t \tag{8}$$

where $\lambda > 0$ is the speed of mean reversion and W_t is a standard Wiener process. In discrete time ($\Delta t = 1$ bar), the exact Euler-Maruyama discretization yields an autoregressive AR(1) regression on the spread changes:

$$\Delta S_t = S_t - S_{t-1} = c + \theta S_{t-1} + \xi_t \tag{9}$$

Using our OLS solver with dependent variable ΔS_t and regressor S_{t-1}, we obtain the slope $\hat{\theta}$, corresponding to an autoregressive coefficient $\phi = 1 + \hat{\theta}$. For the discrete AR(1) process to be mean-reverting and strictly stationary, we require $0 < \phi < 1$. The exact expected time required for the spread to decay halfway back to its mean—the **Half-Life** (HL)—is given by:

$$HL = \frac{\ln(0.5)}{\ln(1 + \hat{\theta})} \quad \text{(in bars)} \tag{10}$$

For very small values of $\hat{\theta}$, the continuous-time approximation $HL \approx -\ln(2)/\hat{\theta}$ is often used in practice. If $\hat{\theta} \geq 0$, the spread is divergent and we assign $HL = \infty$.

2.4 Error Correction Model (ECM) and Bidirectional Feedback

Passing the ADF test confirms that ε_t is stationary, but it does not tell us *which* asset actually does the work of correcting the equilibrium error. Since we structure our regression with the larger-cap stock X as the explanatory benchmark and the smaller-cap stock Y as the follower, we want to ensure that Y actively responds to the disequilibrium.

By the Granger Representation Theorem, any cointegrated system has an Error Correction Model (ECM) representation. We estimate two separate linear regressions on the first differences Δy_t and Δx_t against the lagged disequilibrium error $\hat{\varepsilon}_{t-1} = y_{t-1} - \hat{\alpha} - \hat{\beta} x_{t-1}$:

$$\Delta y_t = \mu_y + \alpha_y \hat{\varepsilon}_{t-1} + u_{y,t} \tag{11}$$

$$\Delta x_t = \mu_x + \alpha_x \hat{\varepsilon}_{t-1} + u_{x,t} \tag{12}$$

Using $N_{\text{obs}} = T - 1$ observations and $df = N_{\text{obs}} - 2$ degrees of freedom, we compute the exact t-statistics for the adjustment coefficients α_y and α_x from the covariance matrix $s^2(\mathbf{W}^T\mathbf{W})^{-1}$. For a pair to be accepted as a valid ECM mechanism, we impose two conditions at significance level $\alpha_{\text{sig}} = 0.05$:

1. **Active Pullback on Leg Y:** When the spread is positive ($\hat{\varepsilon}_{t-1} > 0$, meaning Y is overpriced relative to X), Y must subsequently fall ($\Delta y_t < 0$). Thus, we test $H_1 : \alpha_y < 0$ via a one-sided left-tailed t-test and require $\hat{\alpha}_y < 0$ with $p_y < 0.05$.

2. **No Destructive Feedback on Leg X:** If $\hat{\varepsilon}_{t-1} > 0$, the benchmark X should either remain weakly exogenous ($\alpha_x \approx 0$) or rise to close the gap ($\alpha_x > 0$). If $\hat{\alpha}_x < 0$ and is statistically significant ($p_x < 0.05$ in a two-sided test), then X is falling faster than Y and actively tearing the spread apart. Such pairs are rejected.

2.5 Multivariate Cointegration: The Johansen Trace Procedure

The Engle-Granger method has two well-known limitations: it is restricted to $N = 2$ assets, and the OLS estimate depends on the arbitrary choice of which variable is placed on the left-hand side. To trade N-legged baskets (specifically $N = 3$ triplets in our implementation), we employ the Johansen maximum likelihood framework based on a Vector Error Correction Model (VECM).

Let $\mathbf{P}_t \in \mathbb{R}^{N\times 1}$ be a vector of N log-prices at time t. A VAR(k) model in levels can be rewritten in VECM form with $k-1$ lagged differences (in our code, `k_ar_diff = 1`, corresponding to VAR(2)):

$$\Delta\mathbf{P}_t = \mathbf{\Pi}\mathbf{P}_{t-1} + \mathbf{\Gamma}_1\Delta\mathbf{P}_{t-1} + \mathbf{\Phi}\mathbf{D}_t + \eta_t \tag{13}$$

where $\mathbf{\Pi} = \alpha\beta^T$ is an $N \times N$ long-run impact matrix. The rank $r = \text{rank}(\mathbf{\Pi})$ determines the number of independent cointegrating vectors ($0 \le r \le N-1$).

Johansen's procedure solves a generalized eigenvalue problem via canonical correlation analysis between $\Delta\mathbf{P}_t$ and $\mathbf{P}_{t-1}$ (after concentrating out the short-run dynamics), yielding ordered eigenvalues $\hat{\lambda}_1 > \hat{\lambda}_2 > \cdots > \hat{\lambda}_N$ and corresponding eigenvectors $\hat{\mathbf{v}}_i$. To determine the cointegration rank r, we sequentially evaluate the **Trace Statistic**:

$$LR_{\text{trace}}(r_0) = -T\sum_{i=r_0+1}^{N}\ln(1-\hat{\lambda}_i) \tag{14}$$

Starting from $r_0 = 0$, we test $H_0 : r \le r_0$ against $H_1 : r > r_0$. If $LR_{\text{trace}}(0)$ exceeds the critical value (at the 95% confidence level, `significance_level = 1`), we reject H_0 and conclude that at least one cointegrating relationship exists ($r \ge 1$).

We extract the eigenvector corresponding to the largest eigenvalue $\hat{\lambda}_1$, as it represents the linear combination with the strongest canonical correlation with the stationary part of the process. We normalize this vector by its first component ($\tilde{\mathbf{w}} = \hat{\mathbf{v}}_1/\hat{v}_{1,1}$) to obtain the basket weights $\{w_1, w_2, w_3\}$, where $w_1 = 1.0$.

2.6 The Multiple Hypothesis Testing Dilemma (FDR vs. Structural Filtering)

During the research phase of this project, we investigated a critical statistical pitfall: **Multiple Hypothesis Testing**. When scanning a sector of M stocks, we evaluate $\binom{M}{2}$ pairs (or $M(M-1)$ directed permutations). For $M = 9$ stocks per sector, this is 36 combinations per sector (144 across 4 sectors); in a single 36-stock pool, it reaches $\binom{36}{2} = 630$ pairs.

If we apply a standard ADF test at significance level $\alpha = 0.05$ to m independent non-cointegrated random walks, the expected number of false positives (Type I errors, or spurious cointegrations) is $m \times \alpha$. To control the **False Discovery Rate (FDR)**—the expected proportion of false positives among all rejected null hypotheses—we initially implemented the **Benjamini-Hochberg (BH) step-up procedure**:

1. Sort the m obtained p-values in ascending order: $p_{(1)} \le p_{(2)} \le ... \le p_{(m)}$.

2. For a target FDR level q (e.g., $q = 0.05$ or 0.10), find the largest rank k such that:

$$p_{(k)} \leq \frac{k}{m} q \tag{15}$$

3. Reject H_0 for all pairs with rank $i = 1, \ldots, k$.

Why Benjamini-Hochberg Was Excluded from the Production Pipeline: In our preliminary empirical experiments, applying the Benjamini-Hochberg threshold $(k/m)q$ resulted in **zero pairs passing the filter** across entire multi-year windows. For example, with $m = 72$ directed pairs and $q = 0.10$, the top-ranked pair ($k = 1$) required an ADF p-value below $0.10/72 \approx 0.00139$. Even genuinely related banking pairs (such as `USB/JPM` with $p = 0.0353$) were rejected.

There are two mathematical reasons why raw BH correction is overly conservative for financial cointegration scanning:

1. **Low Power of the ADF Test in Finite Samples:** Unit root tests are notorious for having low statistical power against near-integrated alternatives in 1-year windows ($T \approx 252$). Requiring $p < 0.001$ virtually guarantees a Type II error (failing to detect real mean-reverting pairs).

2. **Positive Dependency Among Tests:** The $\binom{M}{2}$ pairs formed from M stocks share the exact same underlying time series and are strongly positively correlated, violating the independence assumption of standard multiplicity corrections.

Our Engineering Solution: Instead of relying on a single ultra-low p-value threshold from one test, we keep the standard $p_{\text{ADF}} \leq 0.05$ threshold and replace the formal BH correction with a **domain-specific, 5-stage structural filtering cascade** (Sector isolation → Return Correlation → OLS β & R^2 bounds → Half-Life bounds → ECM feedback test). While this cascade does not offer a formal statistical guarantee against false discoveries, it acts as a robust empirical heuristic. By demanding that a pair satisfies multiple orthogonal economic and dynamical constraints simultaneously, we significantly reduce the likelihood of trading spurious relationships without completely destroying the recall of genuine trading opportunities.

3 Statistical Decision Pipeline and Backtesting Architecture

3.1 Universe Segmentation and Sector Isolation

To prevent the scanner from discovering spurious cross-industry relationships (e.g., pairing a semiconductor manufacturer with a retail grocery chain), our universe of 36 liquid US equities is strictly partitioned into 4 homogeneous sectors of 9 stocks each:

- **`TECH_SEMI`:** `AAPL, MSFT, GOOGL, AMZN, META, NVDA, AMD, QCOM, TXN`
- **`BANKS_FINANCE`:** `JPM, BAC, WFC, C, GS, MS, USB, PNC, STT`
- **`ENERGY_OIL`:** `XOM, CVX, COP, EOG, SLB, MPC, PSX, VLO, OXY`
- **`CONSUMER_STAPLES`:** `KO, PEP, PG, CL, KMB, WMT, COST, TGT, MDLZ`

Beyond economic soundness, sector partitioning provides a massive computational advantage for the Johansen Basket scanner: evaluating 3-asset triplets across 36 stocks globally requires $\binom{36}{3} = 7,140$ VECM estimations per window, whereas scanning within 4 sectors of 9 stocks requires only $4 \times \binom{9}{3} = 336$ estimations—a **21-fold speedup**.

3.2 The Sequential Filtering Cascade (`MarketScanner` & `BasketScanner`)

Within each training window, candidate pairs and baskets must survive a strict sequence of filters ordered from computationally cheapest to most expensive:

3.2.1 Pairs Trading Filter Cascade

1. **Data Completeness & Market-Cap Orientation:** Any ticker with missing observations in the training window is dropped. For every pair (A, B), the stock with the higher market capitalization is assigned as the independent variable X, and the smaller stock as Y.

2. **Filter 0 — Return Correlation Pre-Screen ($\rho_r \geq \rho_{\min}$):** Using a pre-computed correlation matrix of percentage returns $r_t = (P_t - P_{t-1})/P_{t-1}$, we reject pairs with $\rho(r_Y, r_X) < 0.45$ (`1d`) or 0.40 (`1h`). This ensures both assets react in the same direction to daily market shocks.

3. **Filter 1 — OLS Hedge Ratio Bounds ($\beta \in [0.35, 2.50]$):** We estimate $y_t = \alpha + \beta x_t + \varepsilon_t$. If $\beta < 0$, hedging requires shorting or buying both legs simultaneously, destroying market neutrality. If $\beta < 0.35$ or $\beta > 2.50$, one leg dominates over 75% of the capital allocation, turning the pair into an unhedged directional bet.

4. **Filter 2 — Goodness of Fit ($R^2 \geq 0.60$):** At least 60% of the log-price variance of asset Y must be explained by asset X, filtering out noisy, idiosyncratic spreads.

5. **Filter 3 — Half-Life Bounds ($HL \in [HL_{\min}, HL_{\max}]$):** For daily data (`1d`), we require $HL \in [1.0, 30.0]$ trading days. For hourly data (`1h`), we require $HL \in [3.0, 84.0]$ hourly bars (approximately 0.5 to 12 trading days, given 7 trading hours per US session).

6. **Filter 4 — MacKinnon ADF Stationarity ($p_{\mathbf{ADF}} \leq 0.05$):** Verifies $I(0)$ stationarity of the residual series.

7. **Filter 5 — Error Correction Mechanism (ECM):** Requires statistically significant mean-reverting pullback on leg Y ($\hat{\alpha}_y < 0, p_y < 0.05$) and no spread-widening feedback on leg X.

8. **Ranking & Non-Overlapping Portfolio Selection:** Surviving pairs across all active sectors are pooled and sorted by shortest Half-Life (HL ascending). To select the top K pairs without concentrating risk in a single stock, our greedy selector (`select_diversified_pairs`) picks the highest-ranked pairs subject to the constraint that no ticker appears in more than one active pair (unless fewer than K disjoint pairs exist).

3.2.2 Johansen Basket Filter Cascade

For 3-asset triplets (A, B, C) within each sector:

1. **Pairwise Return Correlation Pre-Screen:** All three pairwise return correlations must satisfy $\min(\rho_{AB}, \rho_{AC}, \rho_{BC}) \geq 0.40$ (`1d`) or 0.35 (`1h`).

2. **Johansen Trace Test:** Must reject $H_0 : r = 0$ at the 95% confidence level (`significance_level = 1`).

3. **Basket Half-Life Verification:** The synthetic spread $S_t = \sum_{i=1}^{3} w_i \ln(P_{i,t})$ constructed from the principal eigenvector must exhibit a finite Half-Life within $[1.0, 30.0]$ days (`1d`) or $[3.0, 84.0]$ hours (`1h`).

4. **Ranking:** Surviving baskets are ranked in descending order by their maximum eigenvalue $\hat{\lambda}_1$ (strongest cointegration force) and filtered via `select_diversified_baskets` to avoid overlapping tickers.

3.3 Walk-Forward Validation and Dynamic Window Padding

Evaluating a trading strategy on the same historical data used to estimate β and select pairs (*In-Sample* testing) creates severe overfitting and look-ahead bias. To simulate real-time deployment, we implement a rolling **Walk-Forward Optimization** scheme:

- **Daily Profile (`1d`):** A rolling 12-month training window ($T_{\text{train}} \approx 252$ bars) followed immediately by a 3-month Out-of-Sample (OOS) trading window ($T_{\text{test}} \approx 63$ bars). The window then steps forward by 3 months, re-scanning the market and updating weights.
- **Hourly Profile (`1h`):** Due to API history limits on intraday data (730 days in Yahoo Finance) and faster intraday regime shifts, we use a 3-month training window ($T_{\text{train}} \approx 440$ hourly bars) followed by a 1-month OOS trading window ($T_{\text{test}} \approx 147$ hourly bars).

Dynamic Rolling Z-Score and Window Padding: During the OOS test window, we do not use the static mean and standard deviation from the training period, because slight structural drifts would permanently lock the signal. Instead, we standardize the OOS spread using a rolling window of length W calibrated to the pair's training Half-Life:

$$W = \text{clip}\left(\lfloor m \cdot HL_{\text{train}} \rfloor,\, W_{\min},\, W_{\max}\right) \tag{16}$$

To prevent same-bar look-ahead bias when computing the rolling moments, the mean and standard deviation at bar t are calculated strictly on the **lagged** spread S_{t-1}:

$$\mu_t = \frac{1}{W}\sum_{k=1}^{W} S_{t-k}, \quad \sigma_t = \sqrt{\frac{1}{W-1}\sum_{k=1}^{W}(S_{t-k}-\mu_t)^2}, \quad Z_t = \frac{S_t - \mu_t}{\sigma_t} \tag{17}$$

Because a rolling window of length W produces `NaN` values for the first W observations, feeding only the raw OOS test slice into the calculator wouldblind the strategy for the first W bars of every quarter. Our pipeline resolves this automatically via `extract_padded_slice`: before computing Z_t, it prepends exactly $W + 5$ bars from the tail end of the training period as a warm-up buffer (*window padding*), computes the rolling Z-Score seamlessly across the boundary, and then trims the output back to the exact OOS timestamps $[t_{\text{start}}, t_{\text{end}}]$.

3.4 Signal Generation State Machine (`SignalGenerator`)

Trading signals are generated by a deterministic finite state machine operating on the OOS series Z_t with three states: `position` $\in \{$`none`, `long`, `short`$\}$.

- **Entry Rules (Crossover Detection):** When `position == 'none'`, a position is opened only upon a clean threshold crossover between $t-1$ and t:
  - **Short Spread Entry:** If $Z_{t-1} < Z_{\text{entry}}$ and $Z_t \geq Z_{\text{entry}}$ (spread is abnormally high; we short asset Y and buy asset X).
  - **Long Spread Entry:** If $Z_{t-1} > -Z_{\text{entry}}$ and $Z_t \leq -Z_{\text{entry}}$ (spread is abnormally low; we buy asset Y and short asset X).
- **Exit Rules:** An open position is closed when the first of four conditions is triggered:
  1. **Take Profit (Mean Reversion):** $Z_t < Z_{\text{exit}}$ for a short position, or $Z_t > -Z_{\text{exit}}$ for a long position.

2. **Stop-Loss (Structural Break Protection):** $|Z_t| > Z_{\text{SL}} = 3.5$, protecting the portfolio when cointegration breaks down due to earnings shocks or M&A news.
3. **Time Stop (Holding Period Limit):** The number of bars in the trade reaches $T_{\max} = \lfloor 3 \cdot HL_{\text{train}} \rfloor$. If the spread has not reverted within 3 half-lives, the thesis is invalidated.
4. **End-of-Window Liquidation (`EOF`):** Any position still open on the final bar of the OOS window is closed before the next Walk-Forward re-estimation.

- **Same-Bar Re-Entry Guard:** A boolean flag (`just_closed`) prevents the state machine from closing a trade and opening a new one on the exact same bar.

Table 1: System Hyperparameters Across Time-Scale Profiles (`config.py`)

Parameter / Mechanism	**Daily Profile (`1d`)**	**Hourly Intraday Profile (`1h`)**
Walk-Forward Training Window	12 months (~ 252 bars)	3 months (~ 440 bars)
Walk-Forward OOS Test Window	3 months (~ 63 bars)	1 month (~ 147 bars)
ADF MacKinnon Max p-value	0.05	0.05
Allowed Half-Life Range $[HL_{\min}, HL_{\max}]$	[1.0, 30.0] days	[3.0, 84.0] hours
OLS Beta Bounds $[\beta_{\min}, \beta_{\max}]$	[0.35, 2.50]	[0.35, 2.50]
Minimum OLS R^2	0.60	0.60
Min Return Correlation (Pairs / Baskets)	0.45 / 0.40	0.40 / 0.35
Rolling Z-Score Multiplier m	$3.0 \times HL$	$2.5 \times HL$
Rolling Z-Score Window Bounds $[W_{\min}, W_{\max}]$	[15, 90] bars	[21, 105] bars
Entry Threshold Z_{entry} (Pairs / Baskets)	$\pm 2.0\sigma$ / $\pm 1.5\sigma$	$\pm 1.75\sigma$ / $\pm 1.5\sigma$
Take-Profit Exit Threshold Z_{exit}	$\pm 0.5\sigma$	$\pm 0.5\sigma$
Stop-Loss Threshold Z_{SL}	$\pm 3.5\sigma$	$\pm 3.5\sigma$
Max Holding Period $T_{\max}$	$\lfloor 3 \cdot HL \rfloor$ bars	$\lfloor 3 \cdot HL \rfloor$ bars

3.5 Execution Engine, Capital Allocation, and Transaction Costs

A major flaw in many academic backtests is assuming that a signal generated from the closing price of bar t can be executed at that exact same closing price P_t^{Close}. In reality, once the bar closes at 16:00 EST and the Z-Score is computed, the market is closed.

3.5.1 Eliminating Look-Ahead in Execution: Next-Open Matching

Our `DataLoader` fetches both `Close` and `Open` price matrices from Yahoo Finance (stripping timezone offsets so both daily and hourly timestamps align cleanly). Signals are computed exclusively on $\ln(P_t^{\text{Close}})$, while actual trade entries and exits are mapped to the **opening price of the subsequent bar** (P_{t+1}^{Open}) via a backward shift `next_open = df_open_prices.shift(-1)`.

3.5.2 Dollar-Weighted Capital Allocation

Given initial portfolio capital $C_0 = \$100,000$ and a total window allocation $a_{\text{base}} = 10\%$ divided equally among K active pairs or baskets ($B_{\text{trade}} = C_0 \cdot a_{\text{base}}/K$):

- **For a 2-Legged Pair** with hedge ratio β:

$$C_Y = \frac{B_{\text{trade}}}{1 + |\beta|}, \quad C_X = |\beta| \cdot C_Y \quad \Longrightarrow \quad C_Y + C_X = B_{\text{trade}} \tag{18}$$

- **For an N-Legged Johansen Basket** with eigenvector weights $\{w_1, \ldots, w_N\}$:

$$C_i = B_{\text{trade}} \cdot \frac{|w_i|}{\sum_{j=1}^{N} |w_j|}, \quad i \in \{1, \ldots, N\} \tag{19}$$

3.5.3 Gross PnL, Market Friction, and Short-Borrow Accounting

For any leg i with allocated capital C_i, entry execution price $P_{i,\text{in}}$, exit execution price $P_{i,\text{out}}$, and position direction $d_i \in \{+1 \text{ (Long)}, -1 \text{ (Short)}\}$:

$$\text{PnL}_{\text{gross}} = \sum_i C_i \cdot d_i \cdot \left(\frac{P_{i,\text{out}} - P_{i,\text{in}}}{P_{i,\text{in}}} \right) \tag{20}$$

We deduct two distinct sources of real-world trading friction from every trade:

1. **Execution Friction (Brokerage Commission + Slippage):** We model a one-way commission of $c_{\text{comm}} = 3.0$ bps (0.03%) and a bid-ask/slippage penalty of $c_{\text{slip}} = 3.0$ bps (0.03%), giving a one-way friction rate $f = 6.0$ bps $= 0.0006$ per leg. It is charged both on the initial notional capital at entry and on the marked-to-market notional value at exit:

$$\text{Cost}_{\text{exec}} = \sum_i f \cdot C_i \left(1 + \frac{P_{i,\text{out}}}{P_{i,\text{in}}} \right) \tag{21}$$

2. **Short-Selling Borrowing Fee:** Whenever a leg is held short ($d_i = -1$), the broker charges an annual stock loan fee $r_{\text{borrow}} = 1.0\%$ (0.01/365 per calendar day). To handle both daily and intraday holding times accurately, we compute the exact fractional holding duration in days from the timestamps:

$$\Delta\tau_{\text{days}} = \frac{t_{\text{close}} - t_{\text{entry}}}{86,400 \text{ seconds}}, \quad \text{Cost}_{\text{borrow}} = \sum_{i:\, d_i = -1} C_i \cdot \Delta\tau_{\text{days}} \cdot \frac{r_{\text{borrow}}}{365} \tag{22}$$

The final net profit of the trade is $\text{PnL}_{\text{net}} = \text{PnL}_{\text{gross}} - \text{Cost}_{\text{exec}} - \text{Cost}_{\text{borrow}}$.

3.6 Portfolio Evaluation Metrics and Methodological Limitations

To evaluate strategy performance across different market regimes and time scales in Section 4, we aggregate all closed trades chronologically by their normalized closing date and construct a daily portfolio equity curve $E_t = C_0 + \sum_{s \le t} \text{PnL}_{\text{net},s}$ over all business days t.

It is important to acknowledge two methodological limitations in this evaluation approach. First, portfolio risk metrics (Sharpe ratio and Maximum Drawdown) are evaluated on a realized-trade basis rather than via full continuous Mark-to-Market (MtM) daily revaluation. While this accurately captures net realized economic performance, it inherently smooths interim unrealized equity drawdowns while positions remain open. Second, because pair and basket allocations are weighted strictly by their statistical cointegration coefficients (β and eigenvectors) rather than simple 1:1 nominal values, the portfolio achieves statistical spread neutrality, but not necessarily strict dollar-neutrality or CAPM beta-neutrality.

The primary metrics monitored include:

- **Total Net PnL ($) and Total Return (%):** Measures absolute net capital growth after all commissions, slippage, and short-borrowing costs.
- **Win Rate (%):** The percentage of closed trades with $\text{PnL}_{\text{net}} > 0$. While mean-reversion strategies typically exhibit high win rates (55%–65%), Win Rate alone can be misleading if occasional Stop-Loss exits are large.
- **Profit Factor (PF):** The ratio of gross profits on winning trades to absolute gross losses on losing trades:

$$PF = \frac{\sum_{\text{PnL}_{\text{net}} > 0} \text{PnL}_{\text{net}}}{\left| \sum_{\text{PnL}_{\text{net}} < 0} \text{PnL}_{\text{net}} \right|} \tag{23}$$

  A Profit Factor above 1.0 indicates positive expectancy; values above 1.5 indicate a robust edge.

- **Maximum Drawdown (MDD%):** Measures the worst peak-to-trough percentage decline of the portfolio equity curve relative to the running High-Water Mark ($HWM_t = \max_{s \le t} E_s$):

$$\text{MDD\%} = \max_t \left| \frac{E_t - HWM_t}{HWM_t} \right| \times 100\% \tag{24}$$

- **Annualized Sharpe Ratio (SR):** Computed from the daily portfolio returns $R_t = \text{PnL}_{\text{net},t}/HWM_{t-1}$ across all business days in the OOS horizon (assuming risk-free rate $r_f = 0$ for market-neutral long-short cash neutrality):

$$SR = \frac{\bar{R}_d}{\sigma_{R_d}} \sqrt{252} \tag{25}$$

  Aggregating returns to the daily portfolio level ensures that running multiple parallel pairs ($K > 1$) does not artificially inflate the annualization factor.

4 Empirical Analysis of Trading Scenarios and Conclusions

To rigorously evaluate the proposed statistical arbitrage framework, we executed a batch of eight distinct out-of-sample (OOS) experimental scenarios (`S1` through `S8`). Each scenario isolates a specific structural hypothesis: sector composition, portfolio concentration vs. diversification, crisis resilience during a bear market, entry threshold sensitivity, and daily (`1d`) vs. intraday (`1h`) execution.

4.1 Experimental Matrix and Master Results Summary

Table 2 presents the complete empirical results across all eight scenarios, explicitly decomposing performance into Gross PnL, total market friction (execution commissions, bid-ask slippage, and short-selling borrowing fees), and Net Realized PnL.

4.2 Detailed Scenario-by-Scenario Analysis

4.2.1 Bivariate Pairs vs. Multivariate Baskets (`S1` and `S8`)

A striking pattern across the entire 5-year daily OOS period (2020–2024) is the structural superiority of 3-legged Johansen baskets over 2-legged Engle-Granger pairs:

- In **Scenario 8 (Johansen Basket Solo across all 4 sectors)**, the top-ranked basket achieved a Gross PnL of **$3,958.20** across 103 trades. After deducting $1,340.12 in commissions, slippage, and borrowing fees, it delivered a Net PnL of **+$2,618.08** (+2.62% return on total capital, or +26.2% on the 10% active trade allocation), a Win Rate of 60.2%, a Profit Factor of 1.51, a Maximum Drawdown of just 1.28%, and an annualized Sharpe Ratio of **0.66 (as visualized in the equity curve comparison in Figure 1 and the full tearsheet in Figure 3)**.

- Why did **Scenario 8** ($+\$2,618.08$) outperform **Scenario 1** ($+\$669.38$), which excluded the Technology sector? Examining the quarterly window logs reveals the exact mechanism: in Q1 2022 (Window 09, onset of the Ukraine war and oil price shock), excluding Technology forced the `S1` scanner to pick the oil refinery basket `CVX-MPC-OXY`, which suffered a structural break and lost $-\$1,387.18$. In `S8`, having access to all four sectors allowed the scanner to select a stronger eigenvector in `AAPL-QCOM-TXN` during Q1 2022, turning a $-\$1,387.18$ loss into a $+\$1,214.81$ profit, while seamlessly rotating back into Energy baskets (`XOM-SLB-OXY`, `XOM-SLB-PSX`) in 2024 when oil stabilized.

Table 2: Summary of Out-of-Sample Performance Metrics Across All Experimental Scenarios

Scenario	Strategy	Int.	Trades	Gross PnL	Costs	Net PnL	Win Rate	PF	Sharpe
S1: Baseline (No Tech)	Pairs (Top 1)	1d	53	-$2,114.68	$707.24	**-$2,821.92**	47.2%	0.43	**-0.92**
S1: Baseline (No Tech)	Basket (Top 1)	1d	100	$1,983.72	$1,314.34	**$669.38**	61.0%	1.11	**0.15**
S2: Diversified (3P / 2B)	Pairs (Top 3)	1d	135	$439.39	$734.45	**-$295.06**	57.0%	0.92	**-0.14**
S2: Diversified (3P / 2B)	Basket (Top 2)	1d	199	$1,513.28	$1,299.83	**$213.45**	60.8%	1.03	**0.07**
S3: Bear Market (2022)	Pairs (Top 1)	1d	12	$238.68	$160.97	**$77.71**	66.7%	1.07	**0.09**
S3: Bear Market (2022)	Basket (Top 1)	1d	22	$1,660.71	$277.25	**$1,383.46**	68.2%	2.66	**1.78**
S4: Tech Sector Only	Pairs (Top 1)	1d	15	$76.06	$201.74	**-$125.68**	66.7%	0.92	**-0.05**
S4: Tech Sector Only	Basket (Top 1)	1d	70	$1,039.47	$934.67	**$104.80**	61.4%	1.02	**0.03**
S5: High Entry (2.5σ)	Pairs (Top 1)	1d	41	-$391.04	$548.97	**-$940.02**	48.8%	0.75	**-0.31**
S6: Intraday Baseline	Pairs (Top 1)	1h	94	-$156.63	$1,151.10	**-$1,307.73**	53.2%	0.66	**-1.08**
S6: Intraday Baseline	Basket (Top 1)	1h	142	$145.00	$1,732.26	**-$1,587.25**	52.1%	0.63	**-1.54**
S7: Fast Half-Life (≤ 20)	Pairs (Top 1)	1h	94	-$156.63	$1,151.10	**-$1,307.73**	53.2%	0.66	**-1.08**
S7: Fast Half-Life (≤ 20)	Basket (Top 1)	1h	142	$145.00	$1,732.26	**-$1,587.25**	52.1%	0.63	**-1.54**
S8: Johansen Basket Solo	Basket (Top 1)	1d	103	$3,958.20	$1,340.12	**$2,618.08**	60.2%	1.51	**0.66**

- Conversely, **Single-Pair Trading (`S1 Pairs`)** performed poorly ($-\$2,821.92$, Sharpe -0.92). A 2-stock spread is highly vulnerable to idiosyncratic single-company shocks (earnings surprises, CEO changes, litigation). In a 3-stock Johansen basket, the linear combination of two counter-balancing legs absorbs firm-specific noise much more effectively.

4.2.2 The Diversification Trade-Off: Alpha Dilution vs. Risk Reduction (`S2`)

In **Scenario 2**, we expanded the portfolio from $K = 1$ to $K = 3$ disjoint pairs and $K = 2$ disjoint baskets per quarter, splitting the 10% capital budget equally among active positions:

- **Positive Effect on Pairs:** Diversifying across 3 pairs transformed the Gross PnL from $-\$2,114.68$ (`S1`) to $+\$439.39$ (`S2`), raised the Win Rate from 47.2% to 57.0%, and slashed Maximum Drawdown fourfold—from 2.82% down to just **0.64%**. When a top-ranked pair experienced a structural break (e.g., `C/PNC` in Q1 2020), profits from the second and third pairs (`SLB/COP` and `KMB/MDLZ`) cushioned the blow.
- **Alpha Dilution in Baskets:** For Johansen baskets, however, forcing the system to trade the 2nd-best disjoint basket alongside the 1st-best basket diluted net performance from $+\$2,618.08$ (`S8`, 103 trades) down to $+\$213.45$ (`S2`, 199 trades). True multivariate cointegration is scarce; the principal eigenvector of the #1 ranked basket carries genuine alpha, whereas the #2 basket frequently hovers near the edge of the significance threshold and generates nearly double the transaction costs ($1,299.83 across 199 trades) for a fraction of the gross return.

4.2.3 Crisis Resilience and Market Neutrality: The 2022 Bear Market (`S3`)

Scenario 3 isolates the calendar year 2022—a severe macroeconomic stress test during which the S&P 500 fell by approximately -19.4% and the NASDAQ-100 plunged over -33% amid aggressive Federal Reserve interest rate hikes.

- Both strategies achieved positive net returns during the crash: Engle-Granger Pairs posted a modest gain of **+$77.71** (66.7% Win Rate), while the Johansen Basket generated **+$1,383.46** net profit across 22 trades, achieving an exceptional Win Rate of **68.2%**, a Profit Factor of **2.66**, a tiny Maximum Drawdown of **0.51%**, and an annualized Sharpe Ratio of **1.78 (see the dedicated bear market performance comparison in Figure 4)**.
- Notably, the scanner rotated into semiconductor and banking triplets (`AAPL-QCOM-TXN` in Q1 2022 for $+\$1,214.81$, and `JPM-BAC-PNC` in Q3–Q4 2022 for $+\$684.81$). Because every position is dollar-neutral (long and short legs balanced by $|\beta|$ or $|w_i|$), the portfolio remained immunized against the broad market sell-off while harvesting the elevated volatility of relative spread oscillations.

4.2.4 Sector Suitability: Why Pure Technology Struggles (`S4`)

When restricting the scanner exclusively to the `TECH_SEMI` sector (**Scenario 4**), Pairs Trading failed to find any cointegrated pairs in **14 out of 20 quarters** (generating only 15 trades over 5 years, Net PnL $-\$125.68$). Even for Johansen Baskets, restricting the universe to Tech reduced Gross PnL nearly fourfold compared to the full universe (from $3,958.20 in `S8` to $1,039.47 in `S4`), leaving a barely break-even Net PnL of $+\$104.80$ (Sharpe 0.03) after $934.67 in costs.

From an economic standpoint, technology and semiconductor firms undergo rapid, asymmetric innovation cycles (e.g., NVIDIA's AI-driven valuation expansion in 2023–2024 vs. Intel or Texas Instruments). Their log-prices exhibit strong idiosyncratic momentum rather than mean-reverting equilibrium, making them structurally inferior to homogeneous commodity or banking sectors (`ENERGY_OIL`, `BANKS_FINANCE`, `CONSUMER_STAPLES`).

4.2.5 Entry Threshold Sensitivity on Daily Pairs (`S5`)

In **Scenario 5**, we tested whether raising the Z-Score entry threshold for single-pair trading from $Z_{\text{entry}} = 2.0\sigma$ to an extreme 2.5σ could rescue the bivariate strategy.

- Compared to `S1 Pairs` (2.0σ), raising the threshold to 2.5σ reduced the trade count from 53 to 41, cut transaction costs from \$707.24 to \$548.97, and reduced net losses by **66.7%** (from $-\$2,821.92$ to $-\$940.02$, improving the Profit Factor from 0.43 to 0.75 and turning quarters such as Q1 2021 `VLO/CVX` and Q2 2022 `AMD/MSFT` strongly profitable).
- Nevertheless, the strategy remained net negative because at 2.5σ, the distance to the Stop-Loss (3.5σ) is only 1.0σ, whereas the distance to the Take-Profit (0.5σ) is 2.0σ. Consequently, any pair undergoing a genuine regime shift quickly breached the 3.5σ Stop-Loss on the next day's opening gap.

4.2.6 The Intraday Microstructure Friction Trap (`S6` and `S7`)

Scenarios 6 and 7 evaluate the system on hourly intraday bars (`1h`) over 17 rolling 1-month OOS windows (May 2025 – September 2026). Both scenarios produced deeply negative net returns: $-\$1,307.73$ for Pairs (Sharpe -1.08) and $-\$1,587.25$ for Baskets (Sharpe -1.54).

The decomposition of Gross vs. Net PnL in Table 2 provides a textbook demonstration of the **Microstructure Friction Trap**:

1. **Positive Gross Alpha Erased by Transaction Costs:** Notice that on 1-hour bars, the Johansen Basket actually generated a **positive Gross PnL of +\$145.00** with a Win Rate of 52.1% across 142 trades. However, executing 142 three-legged basket trades required $142 \times 3 \times 2 = 852$ individual market orders. At 6 bps one-way friction per leg plus short-borrowing fees, total transaction costs reached **\$1,732.26**—more than $11\times$ the gross profit! **The complete cost decomposition across all scenarios is illustrated in Figure 2.**

2. **Volatility Compression Across Time Scales:** Since stock price volatility scales approximately with the square root of time ($\sigma_{\Delta t} \propto \sqrt{\Delta t}$), a 1.5σ spread deviation on a 1-hour chart represents a price move roughly $\sqrt{7} \approx 2.65\times$ smaller than on a daily chart. Meanwhile, brokerage commissions and bid-ask spreads remain fixed in percentage terms (12 bps round-trip). As a result, fixed transaction costs consume over 80% of the expected mean-reversion move on hourly bars.

3. **Invariance to the Fast Half-Life Cap (`S6` vs. `S7`):** Why did **Scenario 7** ($HL \leq 20$ hours) produce the exact same trades and PnL as **Scenario 6** ($HL \leq 84$ hours)? Because our scanner sorts surviving candidates in ascending order of Half-Life (or descending order of λ_1, which is monotonically inversely related to Half-Life). In every single 3-month training window on 1-hour data, the top-ranked pair and top-ranked basket already had an estimated Half-Life well below 20 hours (typically 4–12 hours, i.e., 0.5–1.5 trading sessions). Thus, tightening the upper cap from 84 to 20 hours did not alter the #1 ranked selection in any window.

4.3 Final Conclusions and Key Takeaways

Based on our mathematical modeling and multi-scenario empirical backtesting, we formulate four primary conclusions:

1. **Empirical Advantage of Multivariate Johansen Baskets:** Within our specific evaluated universe and timeframe, 3-asset Johansen baskets exhibited higher Win Rates (60%–68%) and more robust Profit Factors (1.11–2.66) than standard 2-asset Engle-Granger pairs.

The data suggests that adding a third cointegrated leg may help stabilize the stationary relationship against isolated, firm-specific shocks.

2. **Concentration vs. Diversification Trade-Off:** While splitting capital across multiple pairs (`S2`) successfully reduced realized portfolio drawdown, forcing a Johansen strategy to trade lower-ranked baskets diluted the performance of the principal eigenvector and amplified commission drag. In our tests, allocating exclusively to the single strongest basket across the multi-sector universe (`S8`) yielded the highest overall net profit.

3. **Behavior During the 2022 Bear Market:** The tested strategies demonstrated resilience during the 2022 equity downturn (`S3`). The Johansen Basket configuration generated positive net returns (Sharpe 1.78) during this period. While this highlights the potential of relative-value strategies as portfolio diversifiers during broad market sell-offs, further continuous MtM risk analysis would be required to fully confirm its structural market neutrality.

4. **Time-Scale Boundary and Transaction Cost Dominance:** Standard cointegration models estimated via OLS/VECM are viable on daily intervals (`1d`) where expected price convergence exceeds 2%–4% per trade. On hourly intraday intervals (`1h`), the compression of spread amplitude relative to fixed brokerage commissions and bid-ask slippage (6 bps per leg) creates an insurmountable friction barrier, turning positive gross alpha (+\$145.00) into severe net losses ($-\$1,587.25$). Profitable intraday statistical arbitrage requires sub-basis-point institutional fee tiers and limit-order book execution rather than market-order *Next-Open* execution.

A Supplementary Figures and Performance Visualizations

The following figures provide visual validation for the quantitative performance metrics, scenario comparisons, and walk-forward backtesting architectures discussed throughout the report.

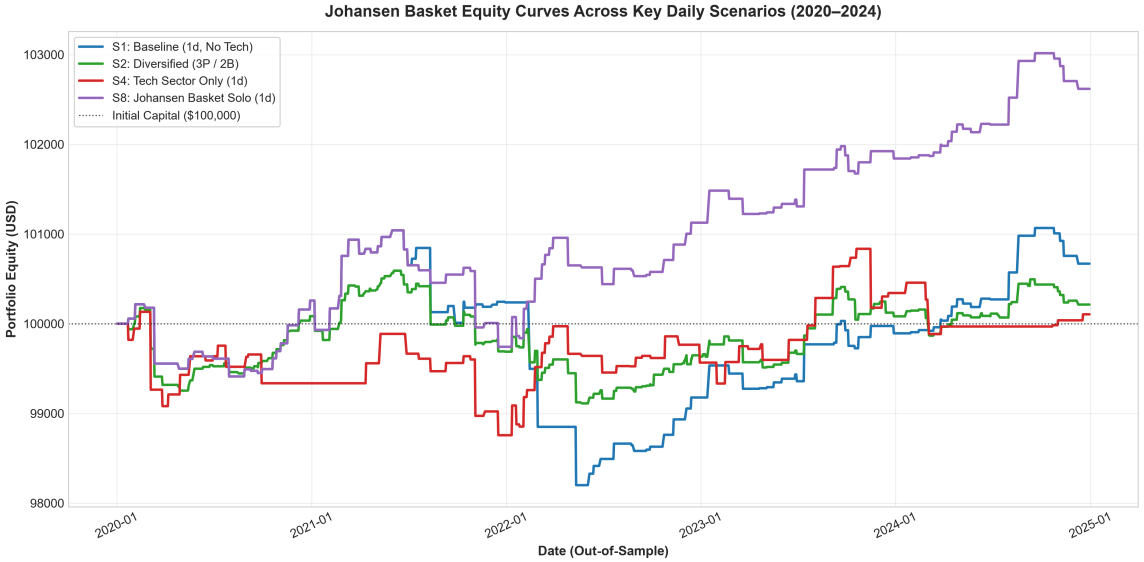


Figure 1: Comparison of Out-of-Sample Portfolio Equity Curves for Johansen Baskets across key daily scenarios (2020–2024).

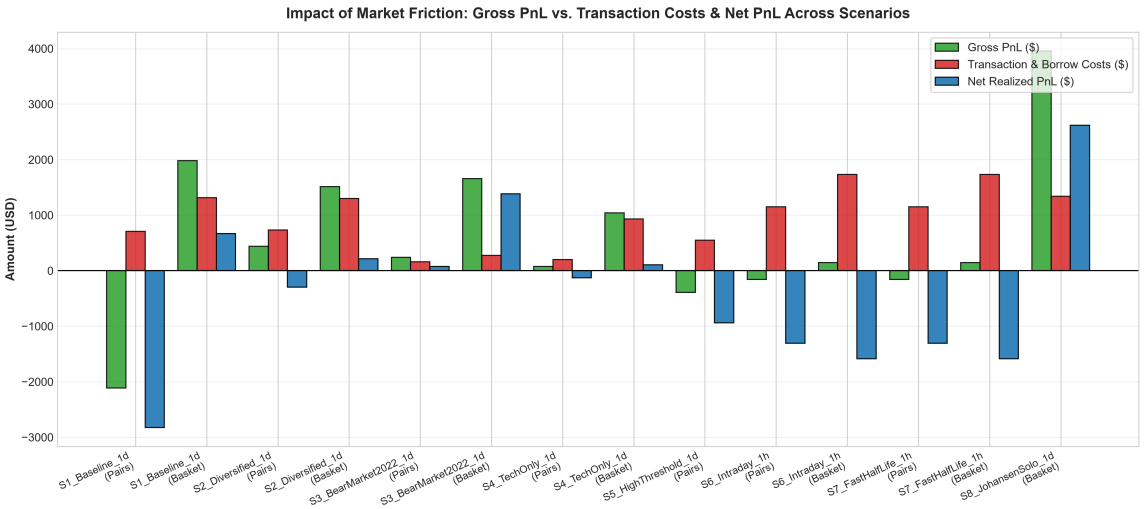


Figure 2: Decomposition of Gross PnL, Transaction & Borrowing Costs, and Net Realized PnL across all experimental scenarios, illustrating the intraday friction trap in S6/S7.

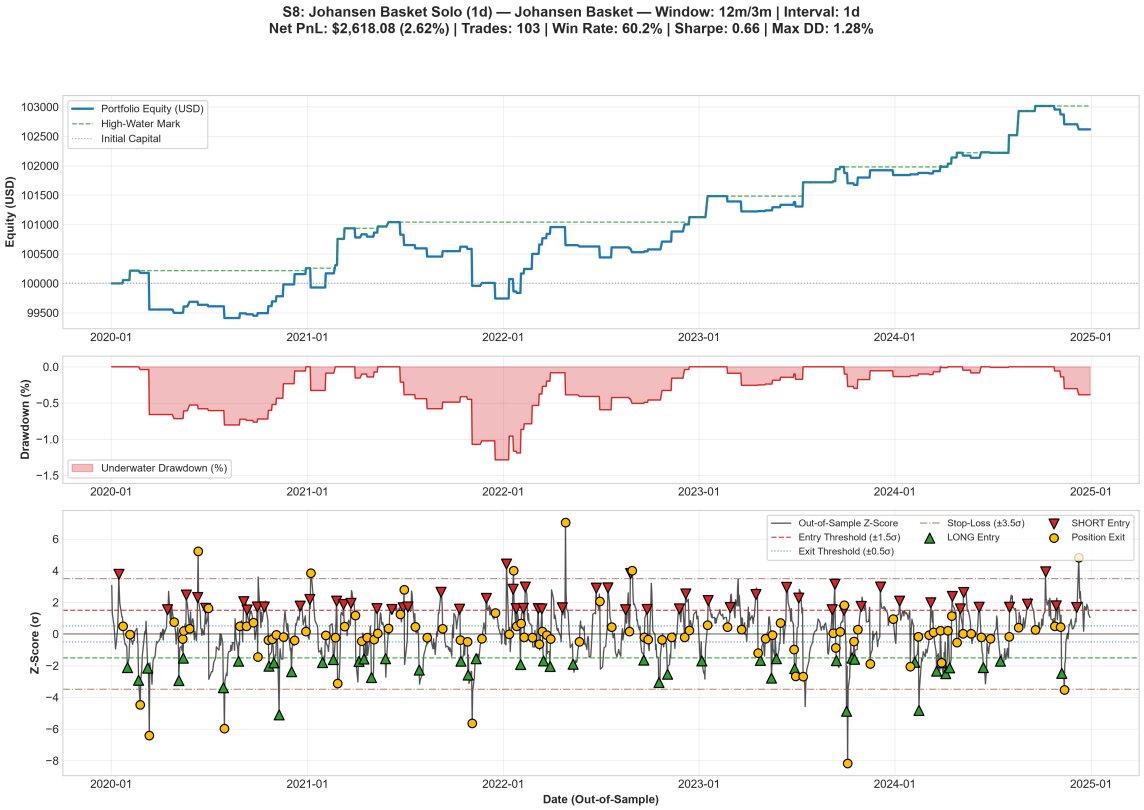


Figure 3: Three-panel Walk-Forward Tearsheet (Equity Curve, Underwater Drawdown, and Z-Score Signals) for Scenario 8 (Johansen Basket Solo, 1d, 2020–2024).

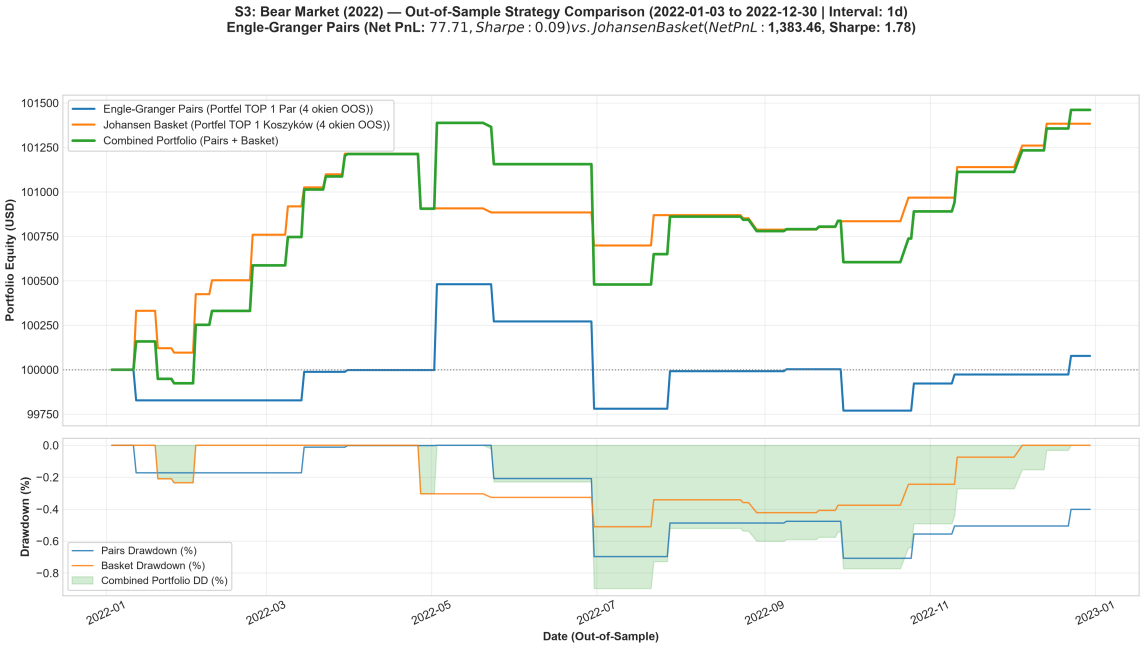


Figure 4: Out-of-Sample Strategy Comparison during the 2022 Bear Market (Scenario 3), demonstrating strong crisis alpha and market neutrality.